Chapter 2: Generative AI

Learning Objectives

By the end of this chapter, you will:

- Understand what Generative AI is and how it differs from traditional AI systems.
- Identify common enterprise use cases and implementation patterns for Generative AI.
- Learn the responsibilities of an AI Technical Program Manager when delivering Generative AI initiatives.

Beginner Explanation

Generative AI (GenAI) refers to artificial intelligence systems that can create new content such as:

- Text
- Images
- Audio
- Video
- Code
- Documents

Traditional AI is designed to classify or predict information, while Generative AI creates something entirely new based on patterns learned from large datasets.

Examples

Application	Traditional AI	Generative AI
Customer Support	Classify tickets	Generate responses
Software Development	Detect bugs	Generate code
Marketing	Analyze campaigns	Create content
HR	Screen resumes	Draft job descriptions

Think of Generative AI as an extremely knowledgeable assistant that can help people work faster. However, it is not always correct and must be governed appropriately.

Real-World Examples

- ChatGPT writing an email.
- GitHub Copilot generating code.
- AI creating meeting summaries.
- AI generating product images.
- AI building PowerPoint presentations.

As an AI TPM, your role is not to build the model—it is to ensure the solution delivers measurable business value.

Intermediate Explanation

Most enterprise Generative AI projects follow a common pattern:

1. Identify a business problem.
2. Select an AI model.
3. Integrate enterprise data.
4. Establish governance controls.
5. Deploy and monitor.
6. Measure business outcomes.

Enterprise Adoption Areas

Department	Use Case
Finance	Invoice processing
HR	Policy assistants
IT	Helpdesk chatbots
Sales	Proposal generation
Legal	Contract summarization
Operations	Knowledge assistants

Enterprise Challenges

- Hallucinations
- Security concerns
- Data privacy
- Regulatory compliance
- Cost management
- User adoption

A successful AI TPM balances four dimensions:

1. Business Value
2. Technical Feasibility
3. Risk Management
4. Cost Optimization

Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Ensures AI initiatives meet timelines and business objectives.
Risk	Identifies privacy, compliance, and hallucination risks.
Stakeholder Management	Aligns executives, engineering teams, and business units.
Budget	Monitors token usage, infrastructure costs, and ROI.

Enterprise Example

Business Problem

A multinational insurance company receives approximately 25,000 customer emails per month. Support teams spend significant time reading, categorizing, and responding to inquiries.

Solution

Build a Generative AI Customer Support Assistant using Azure OpenAI to:

- Summarize emails.
- Categorize customer issues.
- Generate draft responses.
- Recommend next actions.

Architecture

Components:

- Azure OpenAI
- Azure AI Search
- Azure Functions
- Azure Blob Storage
- Azure Monitor
- Customer Support Portal

Outcome

Metric	Before	After
Average Resolution Time	18 Hours	6 Hours
Manual Effort	High	Reduced by 60%
Customer Satisfaction	78%	91%
Monthly Cost Savings	-	~$45,000

Lessons Learned

- Human approval remains essential.
- Governance should be established before production deployment.
- Token consumption should be monitored weekly.
- Executive sponsorship accelerates adoption.

Azure Implementation

Services Used

Service	Purpose
Azure OpenAI	Generate summaries and responses
Azure AI Search	Retrieve enterprise knowledge
Azure Functions	Workflow orchestration
Azure AI Foundry	Prompt management and evaluations

Implementation Steps

Step 1: Provision Azure OpenAI

Deploy a GPT model in Azure AI Foundry.

Step 2: Configure Enterprise Knowledge

- Create AI Search indexes.
- Upload policy documents.
- Configure embeddings.

Step 3: Build APIs

Use Azure Functions for:

- Request processing
- Authentication
- Logging
- Routing

Step 4: Implement Monitoring

Track:

- Token usage
- Latency
- User satisfaction
- Cost per transaction

Step 5: Establish Governance

Create:

- Approval workflows
- Audit mechanisms
- Access controls
- Incident response plans

Mermaid Diagram

```
graph TD

A[Customer Email]
--> B[AI Processing Layer]

B --> C[Azure OpenAI]

C --> D[Azure AI Search]

D --> E[Azure Functions]

E --> F[Support Portal]

F --> G[Customer Service Team]

G --> H[Customer Resolution]
```

TPM Checklist

- [ ] Define business objectives.
- [ ] Identify key stakeholders.
- [ ] Estimate monthly AI costs.
- [ ] Define KPIs.
- [ ] Conduct security reviews.
- [ ] Obtain legal approval.
- [ ] Establish governance controls.
- [ ] Define rollback procedures.
- [ ] Create executive dashboards.
- [ ] Conduct user training.

Templates

Generative AI Readiness Assessment

Question	Status
Is the business problem defined?	Yes/No
Are stakeholders aligned?	Yes/No
Has legal approved the use case?	Yes/No
Is the budget approved?	Yes/No
Are success metrics defined?	Yes/No

Risk Register

Risk	Impact	Mitigation
Hallucinations	High	Human review
Data Leakage	High	RBAC and encryption
Cost Overruns	Medium	Budget alerts
Low Adoption	Medium	Training programs

KPI Dashboard

KPI	Target
Response Time	<5 seconds
User Satisfaction	>90%
Monthly Cost	<$10,000
Accuracy	>95%
Availability	99.9%

Common Mistakes

1. Assuming AI outputs are always correct.
2. Deploying without governance.
3. Ignoring cloud costs.
4. Skipping user acceptance testing.
5. Failing to define measurable KPIs.
6. Treating GenAI as a replacement for employees instead of an augmentation tool.

7. Underestimating change management requirements.

Key Takeaways

- Generative AI creates new content rather than simply analyzing existing data.
- Enterprise adoption requires governance, security, and cost management.
- Azure provides a robust ecosystem for building scalable Generative AI solutions.
- AI TPMs are responsible for aligning business objectives with technical delivery.
- Every Generative AI project should have defined KPIs, risks, and executive sponsorship.
- Human oversight remains a critical component of successful AI deployments.

Footer

---

charan.ai | Page 7